

The Vowel Space Model: A Didactic Introduction

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Companion tutorial to the research article [arXiv:2111.00868](https://arxiv.org/abs/2111.00868) (*A mathematical model of the vowel space*).

Code, data and multimedia: (this repository).

Abstract

This tutorial develops, step by step and with worked numerical examples, the theoretical content of the research article *A mathematical model of the vowel space* ([arXiv:2111.00868](https://arxiv.org/abs/2111.00868)). It is aimed at graduate students and researchers in speech science, acoustics or signal processing who want to understand *why* a two-variable generative model can fill the vowel space (F_1, F_2) and *how* the coordination function links articulatory models to it. Every construction is illustrated by the validated Python code of this repository, so each section can be reproduced by running one script. The research article itself is available on arXiv; it is not part of this repository.

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1 The problem: a many-to-one articulatory–acoustic map

A vowel sound is characterized, to first order, by its two lowest resonances, the formants F_1 and F_2 . These resonances are determined by the *area function* $A(x)$: the cross-sectional area of the vocal tract as a function of the distance x from the glottis ($x = 0$) to the lips ($x = L$, with $L \approx 17.5$ cm for an adult male). The map $A(x) \mapsto (F_1, F_2)$ is

- **many-to-one**: very different tract shapes give nearly the same two formants (a constrictions at the front or at the back of the tract can act alike on F_1);
- **nonlinear**: doubling a constriction area does not double anything in (F_1, F_2) .

This is why acoustic-to-articulatory inversion has no reliable general solution. The article proposes a construct that restores a *bijection*: a two-dimensional family of area functions, indexed by two latent variables, whose image in the (F_1, F_2) plane is the vowel triangle, with a one-to-one correspondence between the family and its acoustic image.

Key idea 1. Instead of fighting the many-to-one relation on real shapes, build the *simplest possible* parametric family of shapes for which the map to (F_1, F_2) is provably one-to-one, then transfer this construction to human-like articulatory models through a single formula, the *coordination function*.

Notation. Vectors are bold ($\mathbf{v}$); $A(x, \theta)$ denotes an area function of the family parameterized by the cycle angle θ ; $\rho \in [0, 1]$ is a radial amplitude; $\mathbf{1}$ is the constant vector equal to 1 (the neutral tube of unit area).

2 The physical anchor: the Schroeder–Ehrenfest relation

Linearize the acoustics around the neutral tube $\mathbf{1}$. Writing the area function as a small perturbation $A(x) = 1 + \epsilon a(x)$, the eigenfrequencies of the closed–open tube shift according to a Rayleigh quotient. Using Ehrenfest’s analogy $\delta f_i / f_i = \delta E_i / E_i$ between the acoustic modes and a quantum system, [Schroeder \(1967\)](#) showed that the first-order frequency shifts are governed by the *odd cosine Fourier coefficients* of the perturbation:

$$\frac{\delta f_i}{f_i} = -\frac{1}{2} a_i, \quad a_1 = \frac{2}{n} \sum_{k=1}^n A_k \cos\left(\frac{\pi(k-\frac{1}{2})}{n}\right), \quad a_2 = \frac{2}{n} \sum_{k=1}^n A_k \cos\left(\frac{3\pi(k-\frac{1}{2})}{n}\right), \quad (1)$$

where a_1 weights the mode shape $v_1 = \cos(\pi x/L)$ and a_2 the mode shape $v_2 = \cos(3\pi x/L)$. Intuition: the first mode has an antinode at the lips and a node at mid-tube, so a front narrowing (large v_1 positive region) lowers F_1 ; the second mode changes sign twice and is mostly sensitive to the middle third of the tract, controlling F_2 .

Remark 1. Relation (1) is *exact at first order* around the neutral tube only. The whole article can be read as the study of how far this linear rule survives (i) far from the neutral tube, (ii) after a rectification of the area function, and (iii) in articulated models. The answer, derived in §7, is: remarkably far, up to one quadratic term on F_1 .

3 The three-phase mixing function

3.1 From the Gibbs triangle to a cycle

In chemistry, a ternary mixture of components (i, j, k) with proportions summing to one is drawn inside a triangle (the Gibbs triangle); the mixture is the convex combination $p = \alpha i + \beta j + \gamma k$ with $\alpha + \beta + \gamma = 1$. The article uses a different slice of the same construction: fix the *sum* of the three weights and let them oscillate, which produces a *closed cycle* through the three components instead of a filled triangle:

$$P(\theta) = \frac{1}{3} (i + j + k) + \frac{2}{3} \left(i \cos\left(\theta - \frac{\pi}{3}\right) + j \cos(\theta - \pi) + k \cos\left(\theta - \frac{5\pi}{3}\right) \right). \quad (2)$$

Each component is visited at its own phase, $P(\frac{\pi}{3}) = i$, $P(\pi) = j$, $P(\frac{5\pi}{3}) = k$, and the mean over the cycle is the neutral point $\Omega = \frac{1}{3}(i + j + k)$. The acoustic space is a triangle of extreme vowels [u, a, i] with the neutral vowel [ə] at its center; the shape space is a circle through the three seed shapes, with the neutral tube at its center. This triangle $\leftrightarrow$ circle pairing is the geometric core of the model.

3.2 The seed vectors $\{i, j, k\}$

For the *generic model*, the seeds are the simplest vector functions that can push F_1 and F_2 to their extremes through relation (1):

$$i = 1 + v_1 + v_2, \quad j = 1 - 2v_1, \quad k = 1 + v_1 - v_2, \quad (3)$$

with $v_1 = \cos(\pi x/L)$ and $v_2 = \cos(3\pi x/L)$. They satisfy $\Omega = \frac{1}{3}(i + j + k) = 1$ (the neutral tube). Expanding (2) with these seeds gives the *concise vowel equation*:

$$P(\theta) = 1 + a_1(\theta) v_1 + a_2(\theta) v_2, \quad a_1(\theta) = 2 \cos \theta, \quad a_2(\theta) = \frac{4}{3} \sin \frac{\pi}{3} \sin \theta. \quad (4)$$

Thus the two mixing coordinates are nothing but the two Schroeder–Ehrenfest coefficients: *the latent variables of the model are the physically meaningful ones*.

Exercise 1. Show from (2) that $P(\theta + \pi) = 2\Omega - P(\theta)$ (point symmetry about Ω). Deduce that the coefficients of the vowels [ɛ, ɪ, ɔ] are the negatives of those of [u, a, i]: $a_i(\theta + \pi) = -a_i(\theta)$. This *antisymmetry* halves the number of seed shapes to specify.

4 The generic model and its eight vowels

Area functions must be positive; the raw mixture $P(\theta)$ is not. A single explicit nonlinearity is introduced, the *soft rectifier*

$$\text{expif}(y) = \begin{cases} e^{y-1}, & y < 1 \\ y, & y \geq 1 \end{cases} \quad (\text{expif in vsmodel/models.py}), \quad (5)$$

which is C^1 at $y = 1$ and equal to the identity above it. The generic model is then

$$A(x, \theta) = \text{expif}\left(1 + a_1(\theta) \cos \frac{\pi x}{L} + a_2(\theta) \cos \frac{3\pi x}{L}\right), \quad (6)$$

with $L = 17.5$ cm as the only static parameter and $\theta \in [0, 2\pi)$ as the single control. The eight cardinal vowels correspond to eight phases, i.e. eight pairs (a_1, a_2) :

vowel	i	u	o	ɔ	a	ɛ	e	ɪ
θ/π	0	1/3	1/2	2/3	1	4/3	3/2	5/3
(a_1, a_2)	(2, 0)	(1, 1)	(0, α)	(-1, 1)	(-2, 0)	(-1, -1)	(0, $-\alpha$)	(1, -1)
F_1 (Hz)	188	259	377	576	886	586	386	267
F_2 (Hz)	1665	984	848	953	1538	1961	2182	2124

Table 1: The eight cardinal vowels of the generic model ($\rho = 1$, $L = 17.5$ cm), with formants computed by the validated Python TLM (`scripts/simu_gen.py`; values reproduce the article within 1 Hz). $\alpha = \frac{4}{3} \sin \frac{\pi}{3}$. Extrema: min F_1 (i), max F_1 (a), min F_2 (o), max F_2 (e) fall exactly on the four expected vowels.

Sampling $A(x, \theta)$ into $n \geq 100$ tubelets and feeding it to the transmission-line model (TLM) of [Badin and Fant \(1984\)](#) yields the vowel loop of Fig. 1: despite the rectification nonlinearity (5), the circle (a_1, a_2) is mapped to a rounded *triangle* whose corners are the extreme vowels — the acoustic vowel triangle emerges from a circular control parameter.

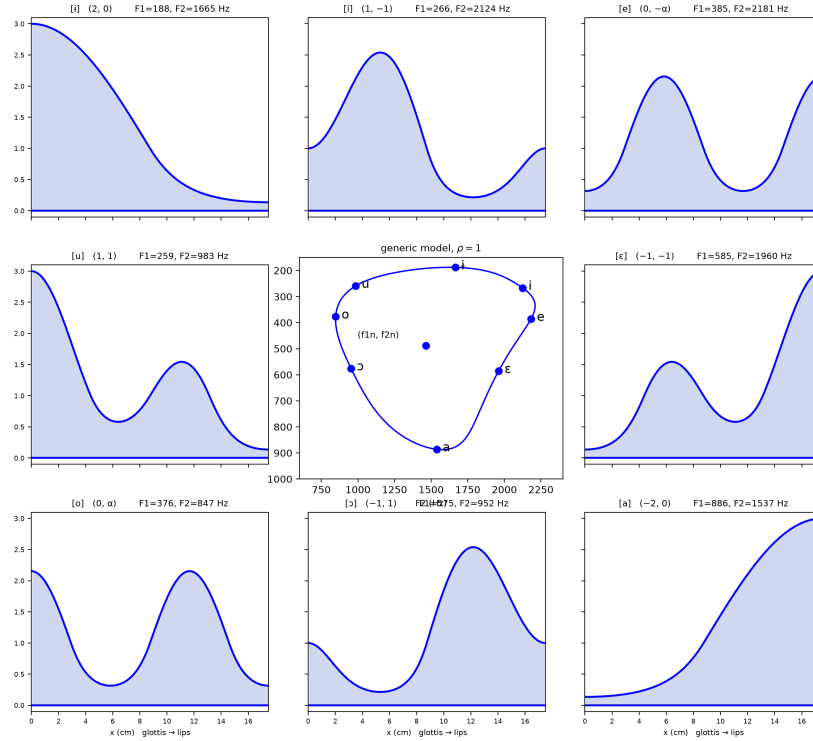


Figure 1: Vowel space of the generic model and the eight area functions (pairs (a_1, a_2)), reproduced by `scripts/fig1_generic.py`. The same construction drives the multimedia files `Mm. 1–2` of the article (`multimedia/generated/MM1.wav`, `MM2.gif`).

Exercise 2. (`scripts/simu_gen.py -quick`) Check that the neutral tube ($\rho = 0$) gives $(F_1, F_2) = (488, 1464)$ Hz and that scaling both coefficients by $\rho \in (0, 1]$ sweeps a homothetic copy of the loop — the domain is filled, not just its boundary.

Remark 2 (Why $n \geq 100$ tubelets?). The sampled staircase approximation of $A(x)$ introduces small shifts of the resonances; with $n = 100$ they stay below ~ 1 Hz for (F_1, F_2) , far below the just-noticeable difference. The article’s results do not depend on the particular TLM implementation, only on n being large.

5 The coordination function

Equation (4) is a special case of a *polar* representation. For *any* vector parameter $\mathbf{p}$ of an articulatory model (section areas, constriction position, tract length, Maeda-style parameters. . .), define the cycle

$$P(\rho, \theta) = \Omega + \rho \Psi_1 \cos(\Psi_2 - \theta), \quad (7)$$

where the three dual vectors are obtained from the seed values $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ of the parameter (its values on the three extreme vowels) by

$$\Omega = \frac{1}{3}(\mathbf{i} + \mathbf{j} + \mathbf{k}), \quad \Psi_2 = \arctan \frac{(\mathbf{i} - \mathbf{k}) \sin \frac{\pi}{3}}{\frac{1}{2}(\mathbf{i} + \mathbf{k}) - \mathbf{j}}, \quad (8)$$

and $\Psi_{1,h} = (h - \Omega) / \cos(\Psi_2 - \theta_h)$ computed componentwise on the first component h that differs from Ω (implemented in `vsmodel/models.py::ijk2psi`). For (a_1, a_2) themselves this gives Table 2, recovering (4): the rewriting is *mathematically equivalent*, but it exposes the *phase locking* of the parameters.

	i	j	k	Ω	Ψ_1	Ψ_2
a_1	1	-2	1	0	2	0
a_2	1	0	-1	0	$\cos^{-1} \frac{\pi}{6}$	$\frac{\pi}{2}$

Table 2: Conversion $\{i, j, k\} \rightarrow \{\Omega, \Psi_1, \Psi_2\}$ for the two Fourier coefficients (Table I of the article). Check with `scripts/fig2_coordination.py`.

Key idea 2. Equation (7) is a *coordination rule*: along the θ cycle, every articulatory parameter oscillates in cosine phase with a fixed phase offset Ψ_2 and amplitude $\rho\Psi_1$ around its offset Ω . Each parameter can be read as *agonist* (in phase with the cycle) or *antagonist* (in opposition) for a given vowel phase — e.g. coding the tract length as $L = \bar{L} + \Delta L \cos(\frac{\pi}{3} - \theta)$ puts the longest tract at the [u] phase, as anatomically expected.

The computational meaning of (7) for 4-tube models is a three-step pipeline, identical for both models below:

1. evaluate the four section values $\mathbf{P}(\rho, \theta) \in \mathbb{R}^4$ from (7);
2. *reconstruct* the sampled n -tube (rounding and merging of section boundaries; geometry differs between models);
3. apply the soft rectifier (5) and compute the formants with the TLM.

6 Two 4-tube models under the same rule

6.1 The DRM: a piston model with an optimal cut

The Distinctive Region Model (Mrayati et al., 1988) cuts the tract at fixed positions $\{L/6, L/2, 5L/6\}$ into four sections of areas $\{A_1, \dots, A_4\}$. The generic model evaluated at the cut-compatible points $x \in \{0, L/3\}$ provides the seeds; because v_1 and v_2 have defined symmetries about the mid-tube, the two back sections follow from the two front ones by *internal antisymmetry*, $P_3 = 2 - P_2$ and $P_4 = 2 - P_1$. With $n = 120$ tubelets split in proportions $\{n/6, n/3, n/3, n/6\}$, the coordination parameters of the first two sections are (Table II of the article)

$$\Omega = (1, 1, 1, 1), \quad \Psi_{2,1} = \arctan\left(\frac{2}{3} \sin \frac{\pi}{3}\right), \quad \Psi_{2,2} = \arctan\left(-\frac{4}{3} \sin \frac{\pi}{3}\right), \quad (9)$$

with $\Psi_1 = \pm(2, 1)/\cos \Psi_2$ and the mirrored values for P_3, P_4 (`vsmodel/models.py::drm_setup`).

6.2 Fant’s model: a sliding constriction

Fant’s four-parameter geometry $\{X_c, A_c, A_l, L\}$ (constriction position and area, lip area, tract length), in the implementation of Badin et al. (1990), has a *constant-length* central section sliding along the tract: the reconstruction step rounds the constriction boundaries, merges the resulting tubelets, then rectifies (Table III of the article; `fant_setup/fant_area`). Here the TLM is run *with* wall-vibration losses, and X_c maximal at the [i] phase builds the large posterior cavity specific to that vowel.

6.3 Conditions C1 and C2: the bijection result

The two models are compared under two Monte-Carlo conditions (`scripts/monte_carlo.py`):

C1 the 4 parameters are drawn independently, uniformly in $\Omega \pm \Psi_1$ (full articulatory potential, no coordination);
C2 (ρ, θ) are drawn uniformly and the parameters are *generated* by (7) (coordinated speech-like configurations).
Figures 3 and 5 of the article (reproduced by `scripts/fig3_drm.py` and `fig5_dfi.py`) show the result: under C1 the cloud of (F_1, F_2) points maps *many* parameter sets to the *same* acoustic point (the scatter in the $(\tilde{a}_1, \tilde{a}_2)$ vs df_i planes is wide), whereas under C2 all points collapse onto two crossing *surfaces* — the map $(\tilde{a}_1, \tilde{a}_2) \leftrightarrow (df_1, df_2)$ becomes one-to-one. The coordination function, not the model, removes the many-to-one relation.

7 How far does the linear rule survive? The quadratic bias

Within the DRM C2 family, the formant deviations from the neutral tube ($df_i = (f_i - f_{in})/f_{in}$) are approximated by a *biased* Schroeder–Ehrenfest relation:

$$d\tilde{f}_1 = -\frac{\tilde{a}_1}{2} - \frac{\tilde{a}_2^2}{4}, \quad d\tilde{f}_2 = -\frac{\tilde{a}_2}{2}, \quad (10)$$

(checked in `scripts/fig6_df_fit.py`). Three remarks for the reader:

- the linear part is exactly (1): the 200-year-old perturbation machinery still sets the scale far from the neutral tube;
- the quadratic term acts on F_1 only; it turns the circle of the linear prediction into the *triangle* of the actual vowel space (Fig. 3 of the article). It is an interference inside the transfer function, not an effect of the rectifier;
- the term is *inherent in the DRM rules* (Table IV of the article): constricting A_1 (or A_3) raises both F_2 -drive and F_1 , which a purely linear rule cannot express.

For Fant’s model, the sliding cut brings the even (symmetric) Fourier components into play, which the formants cannot disentangle — this is the deep source of the many-to-one relation in C1, and the reason the simplest models matter as a baseline.

8 Using the repository

8.1 Layout and quick start

<code>vsmodel/</code>	Python package: TLM (<code>t1m.py</code>), models (<code>models.py</code>), LPC resynthesis (<code>synthesis.py</code>)
<code>scripts/</code>	one script per figure + <code>monte_carlo.py</code> , <code>validate.py</code> , <code>verify_paper.py</code>
<code>data/</code>	reference data converted from the Matlab supplement (<code>data/reference/</code>) and regenerated data
<code>figures/</code>	reproduced Figures 1–7 + validation report
<code>multimedia/</code>	Mm. 1–6: <code>original/</code> (as submitted) and <code>generated/</code> (rebuilt)
<code>matlab_supplement/</code>	original Matlab supplement (SuppPub1–3) and the single-file TLM
<code>paper/</code>	open-access edition <i>source</i> of the article (LaTeX); the article itself is published on arXiv
<code>docs/</code>	this tutorial

Table 3: Repository layout.

Quick start (Python ≥ 3.10 , `numpy` `scipy` `matplotlib`):

```
python scripts/validate.py
python scripts/simu_gen.py (likewise simu_drm.py, simu_fant.py)
python scripts/monte_carlo.py drm (likewise fant)
python scripts/fig1_generic.py ... fig7_overlay.py
python scripts/make_multimedia.py
```

Every script accepts `-quick`; figure scripts accept `-reference` to plot the original Matlab draws. The whole chain is quantitatively checked against the compiled Matlab reference (`figures/validation_report.txt`: formants within ~ 1 Hz, coefficients at machine precision).

8.2 Pointers

- **Research article** (theory, figures, tables): [arXiv:2111.00868](https://arxiv.org/abs/2111.00868) — *A mathematical model of the vowel space*, F. Berthommier. This is the canonical reference; the article PDF is intentionally not hosted in this repository. BibTeX:

```
@article{berthommier2021vowel,  
  title={A mathematical model of the vowel space},  
  author={Berthommier, Frédéric},  
  journal={arXiv preprint arXiv:2111.00868}, year={2021}}
```

(see also CITATION.cff).

- **Multimedia** Mm. 1–6 (vowel sounds, area-function movies): `multimedia/` — regenerate with `scripts/make_multimedia.py`
- **Supplementary Matlab scripts** (as submitted to the journal): `matlab_supplement/` — they plot the figures directly from the included `.mat` reference data.
- **Historical note**: the TLM descends from the transmission-line codes of the Institut de la Communication Parlée (P. Badin, 1984–2006) and the LPC resynthesis from L. Girin; the Python translation is validated against the compiled 2021 binaries that produced the article’s data.

9 Suggested reading path

1. Run `scripts/validate.py` — convince yourself the Python TLM reproduces the reference formants.
2. Read §5 with `scripts/fig2_coordination.py` open: see the two cosines pass through the seed points at $\theta_h \in \{\frac{\pi}{3}, \pi, \frac{5\pi}{3}\}$.
3. Generate the vowel loop (`scripts/simu_gen.py`) and listen to `multimedia/generated/MM1.wav`: the eight phases *are* the eight vowels.
4. Compare C1/C2 (`scripts/monte_carlo.py` then `fig5_dfi.py`): watch the scatter collapse onto the surfaces.
5. Then read the research article [arXiv:2111.00868](https://arxiv.org/abs/2111.00868) for the full discussion (evolutionary perspective, monkey vocal tracts, articulatory potential).

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